

Wavelets and Multiresolution

By
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Ref: Digital Image Processing: Gonzalez and Woods



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INSTRUCTOR

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OBJECTIVE and Components

- **Fourier Transform**
- **Wavelet Transform**
- **Multiresolution And Image Fusion**
- **Image Pyramid and System Block Diagram**
- **Subband Coding**
- **Multilevel Decomposition**
- **Discrete Fourier Transform using Haar Functions.**
- **Wavelets in MATLAB.**
- **Application**

FOURIER TRANSFORM

Fourier Transform decomposes a function of time i.e. a signal, into frequencies that make it up and breaks the signal into an alternative representation characterized by sine and cosines.

→ The Fourier Transform shows that any waveform can be rewritten as the sum of sinusoidal functions.

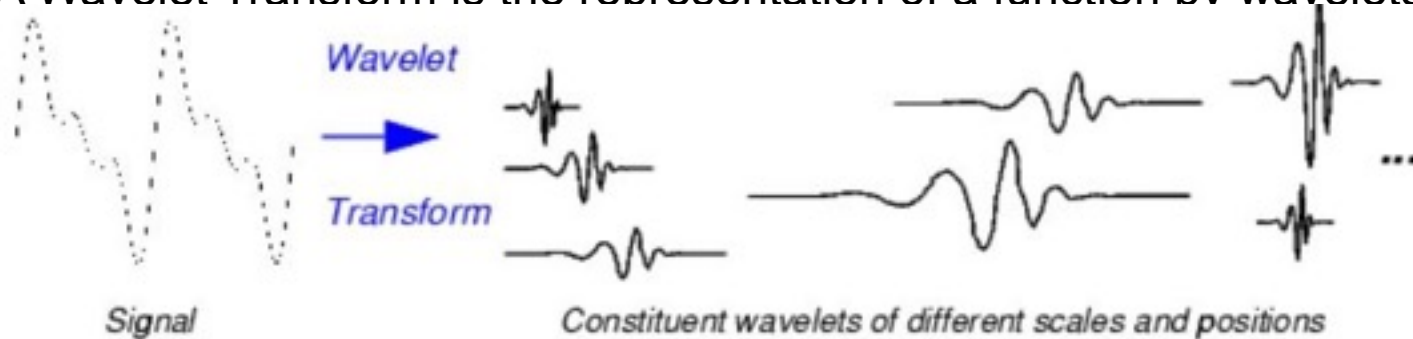
$$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-j\omega t} dt$$



WAVELET TRANSFORM

“The wavelet transform is a tool that cuts up data, functions or operators into different frequency components, and then studies each component with a resolution matched to its scale.”

→ A Wavelet Transform is the representation of a function by wavelets.

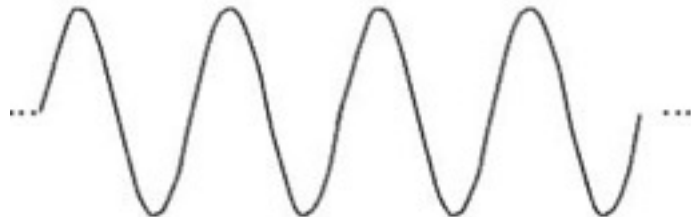


“A wavelet is a mathematical function which is used to divide a given function or continuous time signal into different scale components.



FOURIER VS. WAVELET

- FFT, basis functions: sinusoids
- Wavelet transforms: small waves, called wavelet
- FFT can only offer frequency information
- Wavelet: frequency + temporal information
- Fourier analysis doesn't work well on discontinuous, "bursty" data like music, video, power, earthquakes,...



Sine Wave



Wavelet (db10)

FOURIER VS. WAVELET



Fourier



Loses time (location) coordinate completely



Analyses the whole signal



Short pieces lose “frequency” meaning



Wavelets



Localized time-frequency analysis



Short signal pieces also have significance



Scale = Frequency band

CONTINUOUS WAVELET TRANSFORM

$$CWT_x^{\psi}(\tau, s) = \Psi_x^{\psi}(\tau, s) = \frac{1}{\sqrt{|s|}} \int x(t) \psi^* \left(\frac{t - \tau}{s} \right) dt$$

Continuous wavelet transform is a function of 2 variable -

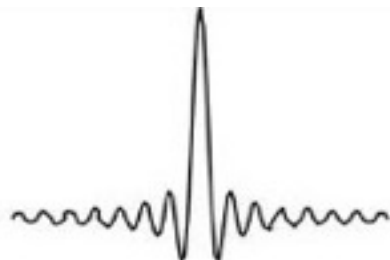
- τ - translation
- S - scale (frequency $^{-1}$)

Where $\psi(t)$ is the transform function, and is called the Mother wavelet.

MOTHER WAVELETS



Haar



Shannon or Sinc



Daubechies 4



Daubechies 20



Gaussian or Spline



Biorthogonal



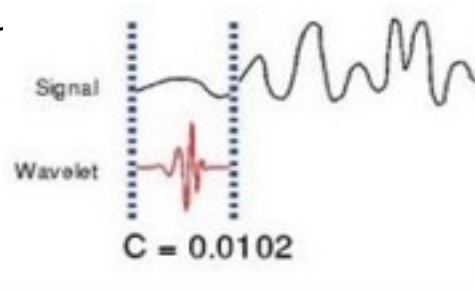
Mexican Hat



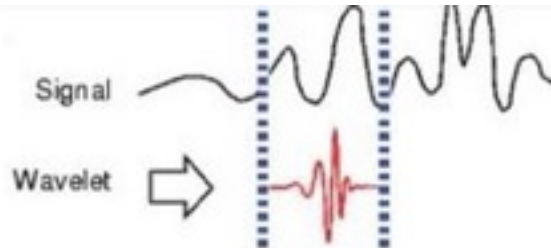
Coiflet

STEPS TO COMPUTE CONT. WAVELET TRANSFORM

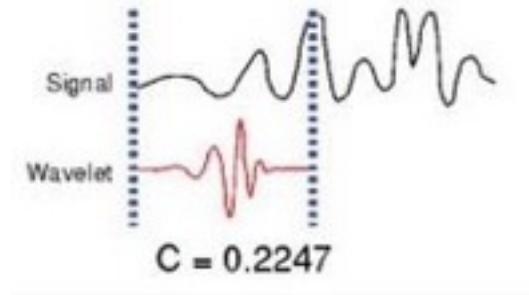
Step 1. Take a mother wavelet and compare it to section at the start of the original signal and calculate a correlation coefficient



Step 2. Shift the wavelet to the right until the whole signal is covered.



Step 3. Scale (stretch) the wavelet and repeat steps 1 through 2.



Step 4. Repeat steps 1 through 3 for all scales.

MULTIRESOLUTION AND IMAGE FUSION

- Normally, we used to work with an image of constant size. But in some occasions, we need to work with images of different resolution of the same image.
- For example, while searching for something in an image, like face, we are not sure at what size the object will be present in the image. In that case, we will need to create a set of images with different resolution and search for object in all the images.
- These set of images with different resolution are called Image Pyramids (because when they are kept in a stack with biggest image at bottom and smallest image at top look like a pyramid)

MULTIRESOLUTION AND IMAGE FUSION

- Multiresolution images in medical researches, represent the combination of different modalities, which generally differ in contrast for different parts of segmented images.
- Modalities refer to the types of images that are created using different image scanning techniques, for example – CT, MRI, etc.
- Multiresolution is needed because no individual sensor is complete, hence, integration of salient features of images produced by different modalities serves for the enhancement of global information.

MULTIRESOLUTION AND IMAGE FUSION

- Multiresolution images are generally created by Fusion technique.
- Image Fusion involves combination of information from different sensors to get much more information compared to the information obtained by a single sensor.
- Most of these approaches for fusion are based on the implementation of the MultiScale Decompositions (MSD).
- Multiresolution is better understood through multi-level image pyramids.

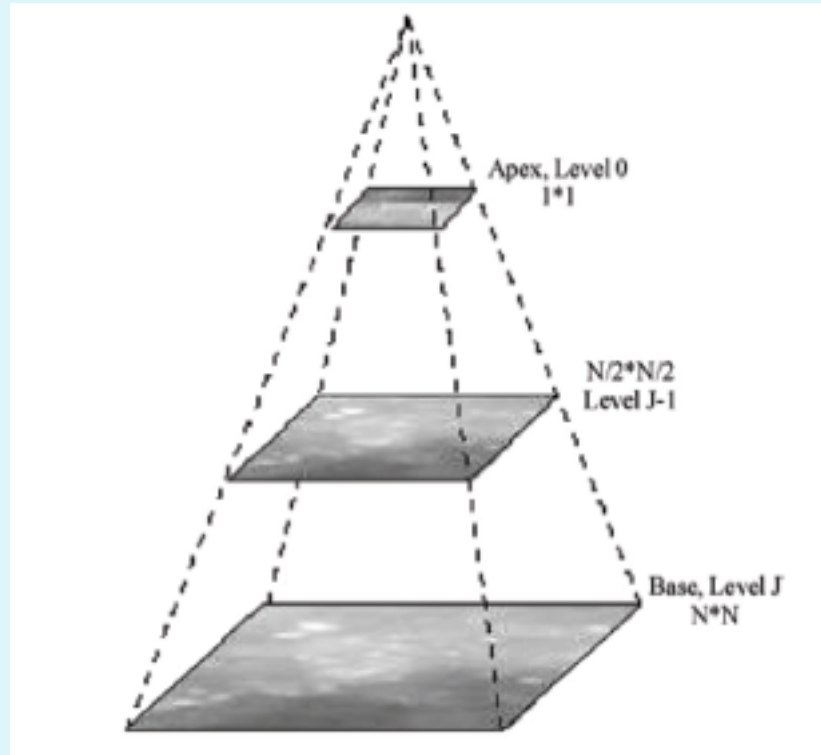
Multiresolution Approach for Decomposition of Input Image in Fine and Coarse Objects

In an image, if both fine and coarse objects or low and high contrast objects are present simultaneously, it is advantageous to study them at several resolutions.

This is the fundamental motivation for multiresolution processing.

The mathematical concept behind this approach is described below.

Image pyramid: An image pyramid is a collection of decreasing resolution images arranged in the shape of a pyramid



The base of the pyramid contains the highest resolution representation of the image.

Moving up the pyramid both size and resolution of the images are decreased.

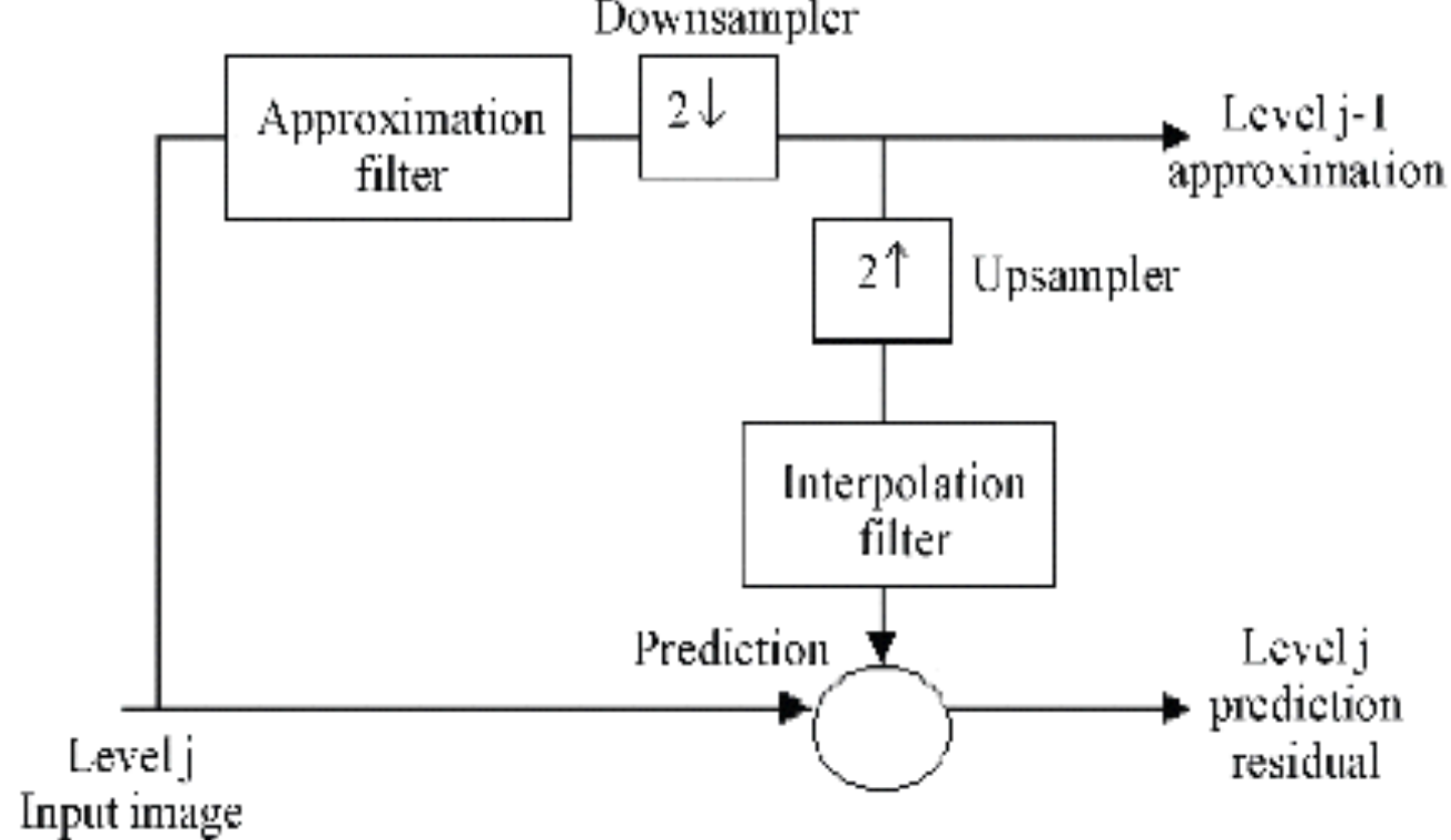
The apex contains the lowest resolution approximation.

The base level J is size $N \times N$ when $N=2^J$, intermediate level j is size $2^j \times 2^j$, where $0 \leq j \leq J$.

Fully populated pyramids are composed of $J+1$ resolution levels from $2^J \times 2^J$ to $2^0 \times 2^0$,

Any image of level j can be approximated to level $j-1$ by applying HAAR wavelet transform.

It will contain only the gross structure of level j . The size of level $j-1$ image is just half of level j image.



To determine the fine details of level j image, we first interpolate the level $j-1$ image to produce one of the same sizes as the level j image. This interpolated image is also known as *prediction of level j image*.

Subtracting the *prediction of level j image* from the original level j image produces level j residual image.

This residual image contains only the fine details or the high frequency sub-band component of level j image.

The technique for producing level $j-1$ approximation and level j prediction residual is the fundamental logic of wavelet based multiresolution processing.

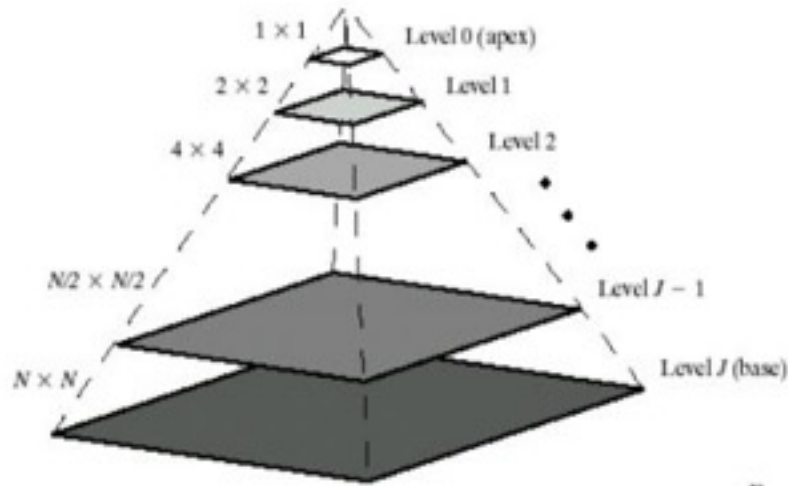
In multiresolution analysis (MRA), scaling function is used to create a series of approximations of an image, each differing by a factor of 2 from its nearest neighbor-ing approximations .

Additional functions, called wave-lets, are then used to encode the difference in information between adjacent approximations.

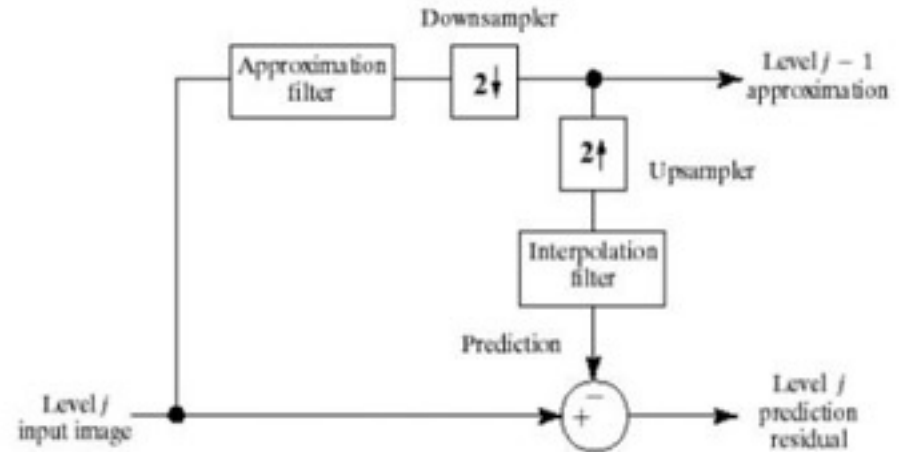
In the multiresolution approach, input image is decomposed into LF and HF subbands by forward wavelet transform whereas after defuzzification, LF and HF subbands are composed by inverse wavelet transform.

IMAGE PYRAMID AND SYSTEM BLOCK DIAGRAM

Total number of elements in a P+1 level pyramid for F

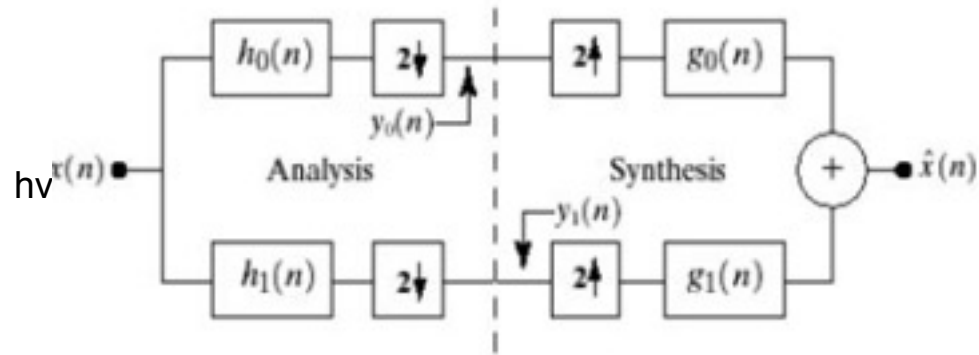
$$N^2 \left(1 + \frac{1}{4} + \frac{1}{4^2} + \dots + \frac{1}{4^P} \right) \leq \frac{4}{3} N^2$$


A pyramidal image structure



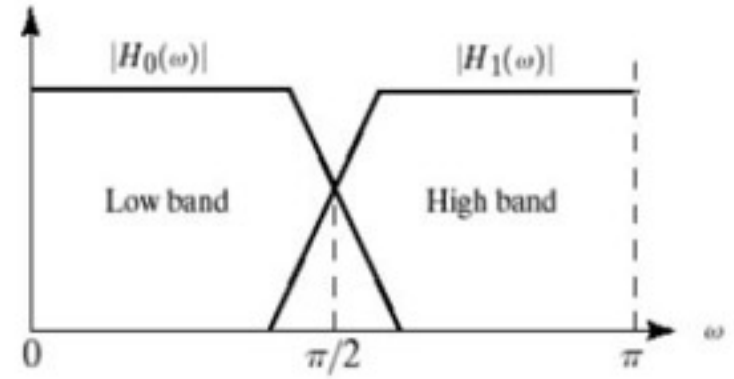
System Block Diagram for

SUBBAND CODING



TWO BAND FILTER BANK

filter

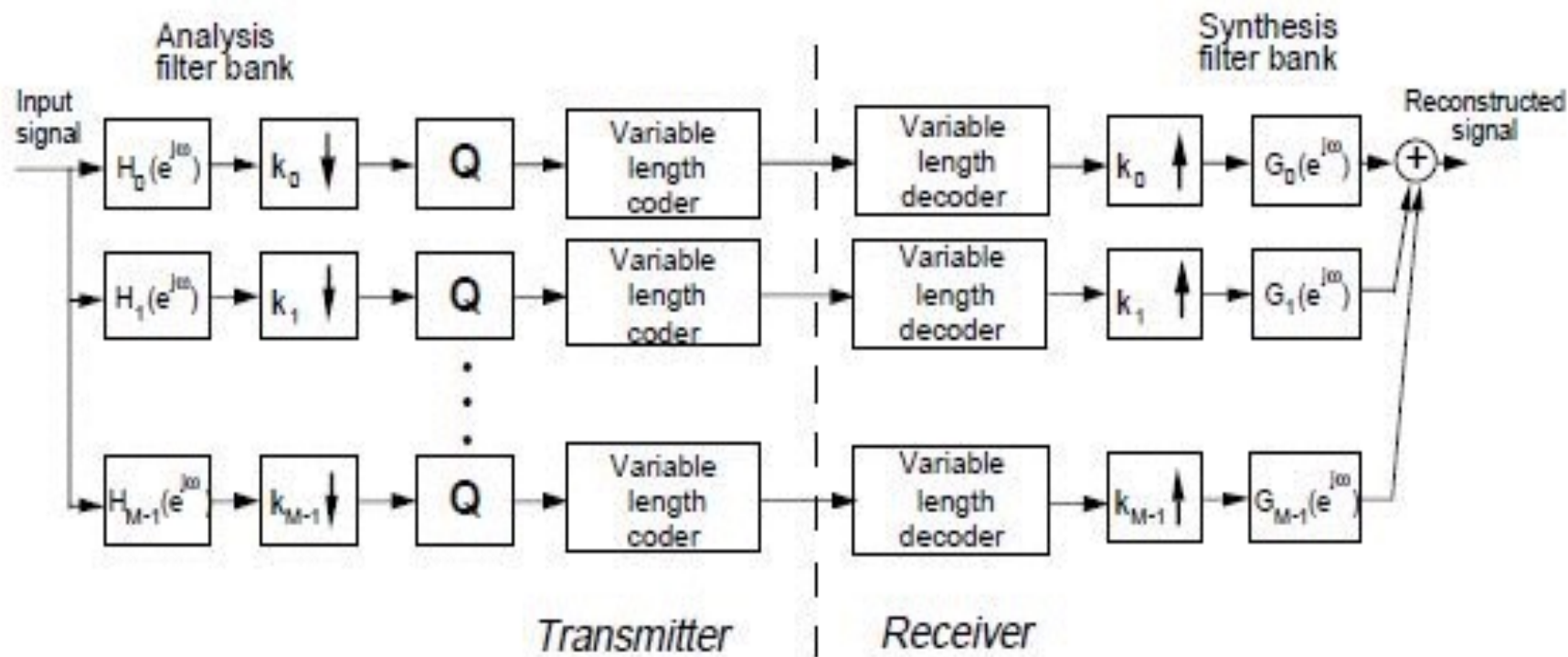


Spectral properties of the bands

- The sampled input signal is convolved in a bank of M bandpass analysis filters with frequency responses $H_i(e^{j\omega})$ and down-sampled by factors k_i that correspond to the bandwidths of $H_i(e^{j\omega})$.
- The frequency responses and down-sampling factors are usually chosen such that the number of samples before and after subband decomposition is the same, i.e.,

$$\frac{1}{k_0} + \frac{1}{k_1} + \dots + \frac{1}{k_{M-1}} = 1.$$

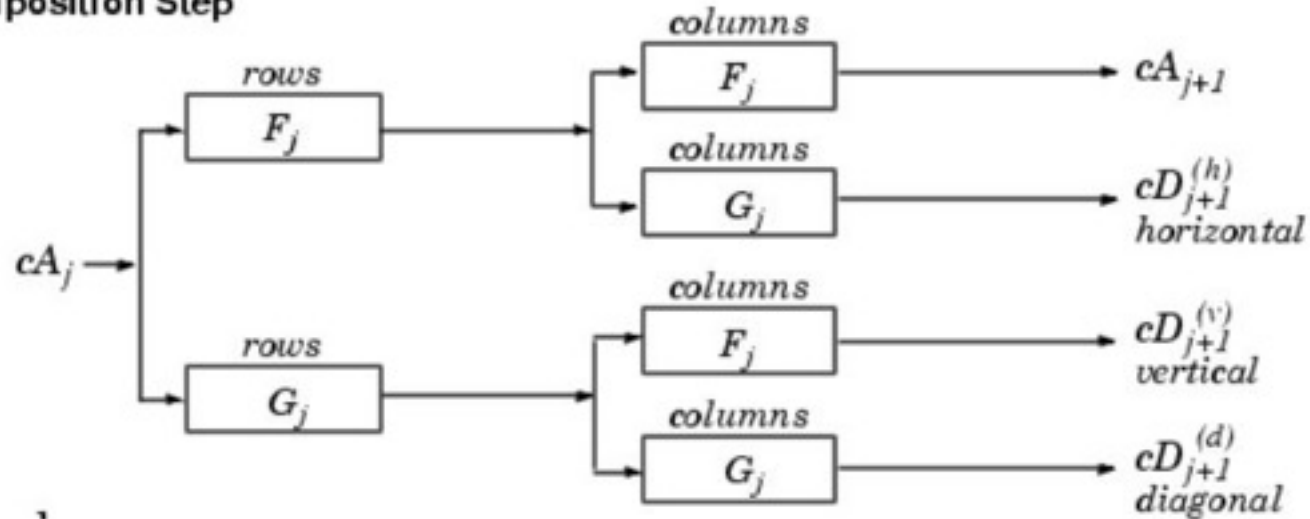
Diagram of a one-dimensional M-channel subband coding scheme.



TWO DIMENSIONAL WAVELET TRANSFORM

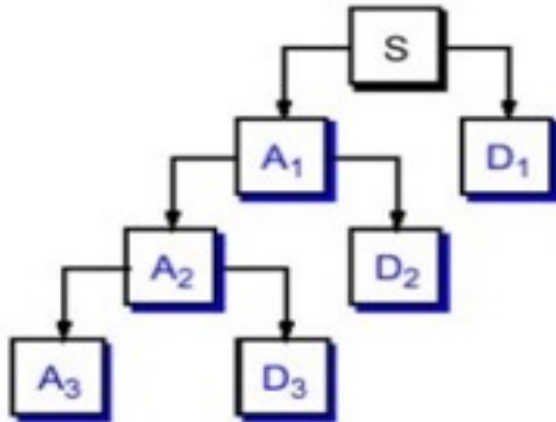
Two-Dimensional SWT

Decomposition Step



MULTILEVEL DECOMPOSITION

Multi-level wavelet decomposition tree



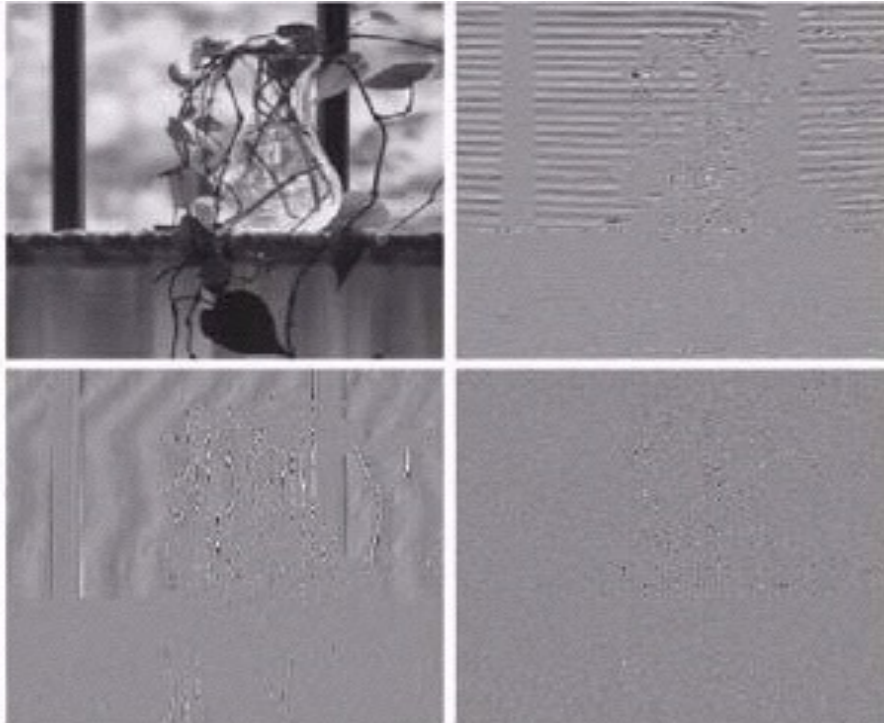
Reassembling original signal

$$S = A_1 + D_1$$

$$= A_2 + D_2 + D_1$$

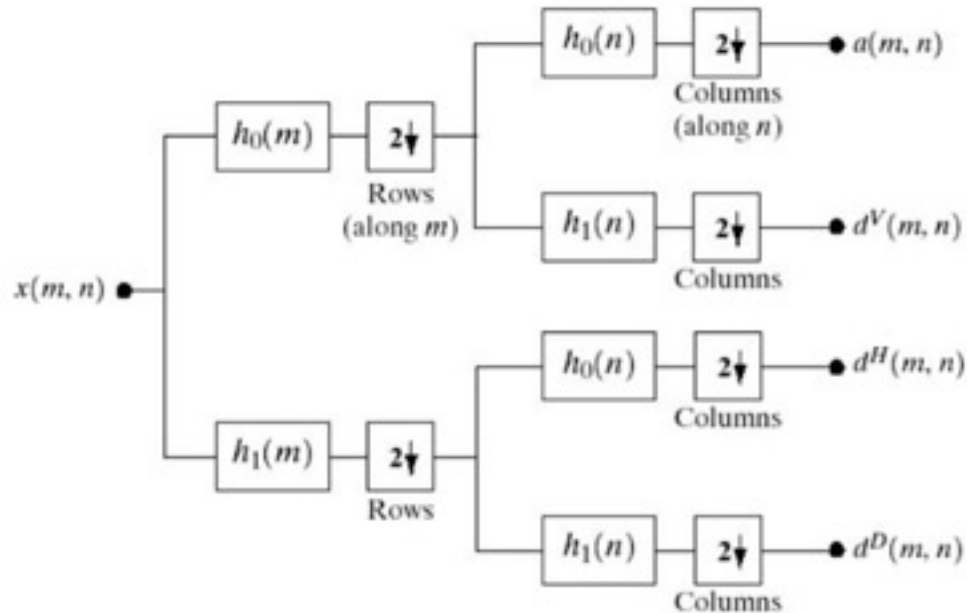
$$= A_3 + D_3 + D_2 + D_1$$

AN EXAMPLE OF DECOMPOSITION (SUBBAND CODING)



The resultant four subbands are

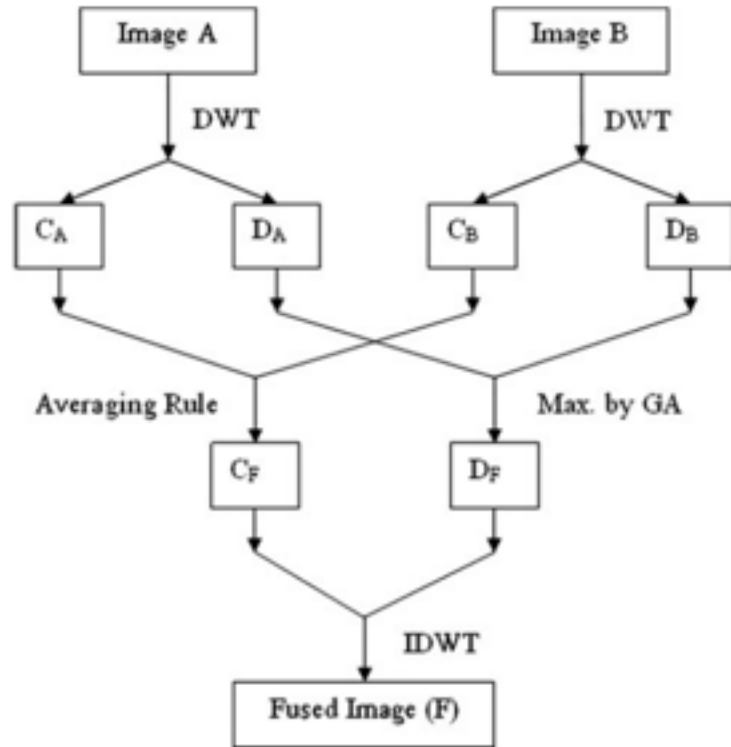
- a) Approximation
- b) Vertical detail
- c) Horizontal detail
- d) Diagonal details



Two-dimensional
four-band filter bank
for subband image
coding.

Most two-dimensional
subband
decompositions are
built by cascading
one-dimensional
subband filter banks.

APPLICATIONS (MULTIRESOLUTION FUSION OF SEGMENTED BRAIN IMAGES)



Proposed Image Fusion Scheme.

- ▀ DWT – Discrete Wavelet Transform.
- ▀ IDWT – Inverse Discrete Wavelet Transform.